

Title of the Project:**Optimal Algebraic Breadth-First Search for Sparse Graphs****Group – G6****Members:**

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Abstract

In this project, we propose modifying and extending the work presented in the paper "Optimal Algebraic Breadth-First Search for Sparse Graphs" by replacing the original BFS-based approach with the GRASP (Greedy Randomized Adaptive Search Procedure) metaheuristic. GRASP, a multi-start randomized greedy algorithm combined with local search, will be adapted to work efficiently on sparse graphs.

We aim to implement GRASP, integrate sparse graph optimization techniques inspired by the reference paper, and analyse its performance in comparison with standard BFS and unoptimized GRASP.

Additionally, a simple Machine Learning or Deep Learning model will be integrated to guide the GRASP algorithm in selecting better next moves dynamically, enhancing search intelligence. A graphical user interface (GUI) will be developed to visualize the graph traversal process, making the demonstration interactive and intuitive. The project includes algorithm design, implementation, analysis, parameter experimentation, and the application of ML/DL methods to improve search performance.

Plane of Work

- **Objective:**

To replace the original BFS-based sparse graph search method with the GRASP (Greedy Randomized Adaptive Search Procedure) metaheuristic, adapt GRASP for sparse graph optimization, analyse its performance, and optionally enhance it using Machine Learning/Deep Learning if time permits.

- **First, analysis of the algorithm:**

GRASP is a multi-start metaheuristic combining greedy randomized construction and local search improvement. It builds solutions step-by-step by selecting candidates from a restricted list based on a greedy function and controlled randomness, allowing better exploration of the solution space. In sparse graphs, GRASP can be optimized by focusing only on a reduced active subset of nodes. Compared to BFS, GRASP naturally handles exploration better, especially in sparse environments.

- **Second, implementation of the algorithm with analysis:**

We will first implement GRASP for sparse graphs. Optimization techniques inspired by the reference paper will be applied to reduce redundant computations. We will compare the performance of BFS, standard GRASP, and optimized GRASP through experiments on sparse graphs. Additionally, if time permits, a simple Machine Learning or Deep Learning model (such as a decision tree or a small neural network) will be trained to predict better next steps during the search to further guide GRASP dynamically.

- **Applications and difficulties:**

GRASP-based methods can be applied in network design, routing, scheduling, and resource optimization problems, especially on large and sparse graphs.

The main challenges include:

- Efficiently incorporating GRASP Algorithm.
- Integrating ML/DL predictions effectively into the search process (if implemented).
- Balancing search diversification and solution quality.
- Visualizing large graphs interactively in the GUI.

Reference paper:

P. Burkhardt, "Optimal Algebraic Breadth-First Search for Sparse Graphs," in *Proceedings of the 26th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP '21)*, pp. 20–31, 2021, doi: [10.1145/3446216](https://doi.org/10.1145/3446216).